

Theory and Experiment on Formation-Containment Control of Multiple Multirotor Unmanned Aerial Vehicle Systems

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Abstract—Formation-containment control problems for multiple multirotor unmanned aerial vehicle (UAV) systems with directed topologies are studied, where the states of leaders form desired formation and the states of followers converge to the convex hull spanned by those of the leaders. First, formation-containment protocols are constructed based on the neighboring information of UAVs. Then, sufficient conditions for multi-UAV systems to achieve formation-containment are presented. An explicit expression to describe the relationship among the states of followers, the time-varying formation for the leaders and the formation reference is derived. It is shown that the states of followers not only converge to the convex hull formed by those of leaders but also keep certain formation specified by the convex combination of the formation for the leaders. Moreover, an approach to determine the gain matrices of the formation-containment protocol is proposed by solving an algebraic Riccati equation. Finally, a formation-containment platform with five quadrotor UAVs is introduced, and both the simulation and experimental results are presented to demonstrate the effectiveness of the obtained results.

Note to Practitioners—This paper addresses the problem of formation-containment control for multi-UAV systems over directed topologies. In practical applications, there may exist multiple leaders and multiple followers in a multi-UAV system. Formation-containment means that the states of leaders form the desired time-varying formation and at the same time the states of the followers converge to the convex hull spanned by those of the leaders. Formation-containment control provides a unified framework for formation control and containment control, and has potential applications in the cooperative source seeking,

load transportation, and surveillance. Although formation control and containment control problems have been studied a lot, the formation-containment control problem for multi-UAV system is still open and challenging. This paper proposed a distributed formation-containment protocol for the multi-UAV system using local neighboring information. Sufficient conditions for multi-UAV systems to achieve formation-containment are presented. It is proven that the states of followers not only converge to the convex hull formed by those of leaders but also keep certain formation specified by the convex combination of the formation for the leaders. An approach to design the formation-containment protocol is given. A remarkable point for this paper is that the obtained results are demonstrated by practical experiments with five quadrotor UAVs.

Index Terms—Cooperative control, formation-containment control, multiagent system, unmanned aerial vehicle (UAV).

I. INTRODUCTION

COOPERATIVE control of multiple unmanned aerial vehicle (UAV) systems has been a hot research topic in recent years due to its broad potential applications in many practical areas, such as source seeking [1], [2], load transportation [3], localization [4], drag reduction [5], surveillance [6], [7], and telecommunication relay [8]. One of the most important issues in cooperative control of multi-UAV systems is formation control which requires the positions of all UAVs to keep a certain geometric configuration. Although some traditional formation strategies that include leader-follower, virtual structure, and behavior-based ones can be applied to deal with the formation control problems of multi-UAV systems (see [9]–[12]), how to design the formation controller using neighboring information is still an attractive topic.

With the development of consensus theory (see [13]–[17] and references therein), more and more researchers realize that consensus strategies can be extended to solve the formation control problems of multiagent systems. Ren [18] studied formation control problems for second-order multiagent systems using consensus-based protocols and proved that leader-follower, virtual structure and behavior-based formation strategies can be viewed as special cases of consensus-based ones. Sufficient conditions for second-order multiagent systems with undirected topologies to achieve time-invariant formations were presented in [19]. Sufficient conditions for first-order and second-order multiagent systems to achieve formation in a finite time were proposed in [20] and [21], respectively. Pursuit formation control problems for second-order multiagent systems were addressed in [22] using consensus

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control approaches. A consensus protocol for second-order multiagent systems was applied to deal with formation control problems of linearized nonholonomic vehicle systems under time-delayed communications in [23]. Abdessameud and Tayebi [24] constructed a formation protocol with time delays for multi-UAV systems modeled by second-order dynamics and proved the stability of the proposed protocol based on the Lyapunov-Krasovskii functional theory. In [25], necessary and sufficient conditions for multi-UAV systems to achieve time-varying formations were presented, and the theoretical results were validated by practical formation experiments. Consensus-based cooperative path following the problems for multi-UAV systems over time-varying networks were investigated in [26]. Time-varying formation control experiments for multi-UAV systems with switching undirected topologies were carried out in [27] and [28].

Besides formation control, another recent hot research topic in cooperative control of multiagent systems is containment, where there exist multiple leaders and the states of followers are required to converge to the convex hull spanned by those of the leaders. A hybrid stop-go control strategy was applied to study the containment problems of multiagent systems in [29]. Finite-time containment problems for multiagent systems with rigid bodies were addressed in [30]. Cao *et al.* [31] demonstrated their theoretical results for containment control of second-order multiagent system on practical mobile robot multiagent systems. Necessary and sufficient conditions for first-order and second-order multiagent systems to achieve containment were proposed in [32]. Containment control problems for second-order multiagent systems without velocity measurements were studied in [33] and [34], respectively. Sufficient conditions for second-order multiagent systems with inherent nonlinear dynamics to achieve containment were presented in [35]. Necessary and sufficient conditions for multi-UAV systems with multiple stationary leaders to achieve containment were derived in [36].

Based on the formation control and the containment control, another more interesting problem is formation-containment control that requires the states of leaders to keep certain formation while the states of followers converge to the convex hull formed by those of the leaders. Formation-containment control has potential applications in the cooperative transportation of multirobot systems across the hazardous areas, where the leader robots with detection devices are required to achieve a given formation configuration and the follower robots can move into the safe region formed by the leaders. Formation-containment control problems for first-order and second-order multiagent systems with undirected topologies were addressed in [37] and [38]. Sufficient conditions for first-order multiagent systems with directed switching topologies were given in [39]. The effects of time delays on the formation-containment control of first-order multiagent systems were discussed in [40]. In practice, undirected topologies mean that the communication among agents is bidirected, which may consume twice the communication and energy resources used in the directed links. Directed topology among UAVs is more practical, considering the limited communication capabilities. To the best of our knowledge, formation-containment control

problems for multi-UAV systems with directed topologies have not been studied extensively.

In this paper, formation-containment problems for multi-UAV systems with directed topologies are investigated. First, formation-containment protocols are constructed using neighboring information of UAVs, where the formation for leaders can be time-varying. Then sufficient conditions for multi-UAV systems to achieve formation-containment are proposed, where a feasible time-varying formation set is given. An explicit expression to describe the relationship among the states of followers, the time-varying formation for the leaders and the formation reference is derived. It is proven that the states of followers not only converge to the convex hull formed by those of leaders but also keep certain formation specified by the convex combination of the formation for the leaders. Moreover, an approach to design the formation-containment protocol is proposed by solving an algebraic Riccati equation. Finally, a formation-containment platform consisting of five quadrotor UAVs is introduced, and both the simulation and experimental results are presented to demonstrate the effectiveness of the obtained results.

Compared with the existing results on formation control, containment control, and formation-containment control of multi-UAV systems, the new contributions of this paper are threefold. First, there exist multiple leaders and multiple followers, and the leaders coordinate with each other to achieve desired formations. In [22], [24], [25], [27], and [28], only formation problems were considered and there exist no multiple leaders. In [36], the leaders are stationary and do not coordinate with each other to achieve formations. The dynamics of each agent in [37]–[40] is limited to first order and cannot be applied to describe the multi-UAV systems in this paper. Second, the formation for leaders is time-varying and an explicit expression to describe the relationship among the states of followers, the time-varying formation for the leaders, and the formation reference are derived. It is proven that the states of followers not only converge to the convex hull formed by those of leaders but also keep certain formation specified by the convex combination of the formation for the leaders. Third, a practical formation-containment experiment is carried out using five quadrotors in the outdoor environment to demonstrate the effectiveness of the obtained results. The containment results in [36] were demonstrated by the numerical simulations.

The rest of this paper is organized as follows. In Section II, the basic concepts and useful results on graph theory are introduced and the problem description is presented. In Section III, sufficient conditions for multi-UAV systems to achieve formation-containment are proposed, and an approach to design the protocol is presented. The simulation and experimental results are provided in Section IV. Finally, Section V concludes this paper.

Throughout this paper, for simplicity of notation, let $\mathbf{0}$ be the zero matrices of appropriate size with zero vectors and zero number as special cases. $\mathbf{1}$ is used to represent a column vector of appropriate size with 1 as its elements. I stands for an identity matrix with appropriate dimension and $\otimes$ denotes the Kronecker product.

II. PRELIMINARIES AND PROBLEM DESCRIPTION

In this section, the basic concepts and the results in graph theory are introduced and the problem description is presented.

A. Basic Concepts and Results in Graph Theory

A directed graph G can be denoted by $G = \{V, \varepsilon, W\}$ with the node set $V = \{v_1, v_2, \dots, v_N\}$, the edge set $\varepsilon \subseteq \{(v_i, v_j) : v_i, v_j \in V, i \neq j\}$, and a weighted adjacency matrix $W = [w_{ij}] \in \mathbb{R}^{N \times N}$ ($w_{ij} \geq 0$). Denote by $e_{ij} = (v_i, v_j)$, the edge of G , where node v_i is called a neighbor of node v_j . The adjacency elements $w_{ij} > 0$ if and only if $e_{ji} \in \varepsilon$, else $w_{ij} = 0$. The neighbor set of node v_i is denoted by $N_i = \{v_j \in V : (v_j, v_i) \in \varepsilon\}$. Define the in-degree of node i as $\deg_{in}(v_i) = \sum_{j=1}^N w_{ij}$. Denote by $D = \text{diag}\{\deg_{in}(v_i), i = 1, 2, \dots, N\}$, the degree matrix of G . The Laplacian matrix of G is defined as $L = D - W$. A directed path from node v_{i_1} to v_{i_k} is a sequence of ordered edges with the form of $(v_{i_l}, v_{i_{l+1}})$, where $v_{i_l} \in V$ ($l = 1, 2, \dots, k-1$). A directed graph is said to have a spanning tree if there exists at least one node having a directed path to all the other nodes. More details on graph theory can be found in [41]. The following lemma is useful in the analysis and design of the formation-containment problems for multi-UAV systems.

Lemma 1 [42]: Let $L \in \mathbb{R}^{N \times N}$ be the Laplacian matrix of a directed graph G . If G has a spanning tree, then 0 is a simple eigenvalue of L with $\mathbf{1}$, the associated right eigenvector, and all the other $N-1$ eigenvalues have positive real parts.

B. Problem Description

For a multi-UAV system with N multirotor UAVs, the interaction topology of the multi-UAV system can be described by a directed graph G with node i representing the UAV i and edge e_{ij} representing the interaction between UAV i and j . The interaction strength from UAV i to UAV j can be described by the adjacency element w_{ji} . On the formation-containment control level, the dynamics of the UAV i , $i \in \{1, 2, \dots, N\}$ can be modeled by the following double-integrator [9], [10], [25], [27]

$$\begin{cases} \dot{x}_i(t) = v_i(t) \\ \dot{v}_i(t) = u_i(t) \end{cases} \quad (1)$$

where $x_i(t) \in \mathbb{R}^n$ is the position vector, $v_i(t) \in \mathbb{R}^n$ is the velocity vector, and $u_i(t) \in \mathbb{R}^n$ is the control input vector. For simplicity of description, n is assumed to be 1, if not otherwise specified. However, all the results hereafter are still valid for higher dimensional space using the Kronecker product.

Remark 1: The dynamics of a multirotor UAV with six degrees of freedom can be classified into trajectory dynamics and attitude dynamics. Due to the fact that the trajectory dynamics have much larger time constants than the attitude dynamics, the control of a multirotor UAV can be implemented with an inner-loop/outer-loop structure (see, e.g., [25], [27], [28], [43], [44], and the reference therein), i.e., the outer-loop drives the UAV toward the desired position, while the

inner-loop tracks the attitude. In this paper, the formation-containment problem for multirotor UAV systems only concerns the positions and velocities. Therefore, the dynamics of the leader and follower UAVs in the outer-loop can be approximately modeled by (1).

Let $\phi_i(t) = [x_i(t), v_i(t)]^T$, $B_1 = [1, 0]^T$, and $B_2 = [0, 1]^T$. Then, multi-UAV system (1) can be rewritten as

$$\dot{\phi}_i(t) = B_1 B_2^T \phi_i(t) + B_2 u_i(t). \quad (2)$$

Definition 1: UAVs in the system (2) are classified into leaders and followers. A UAV is called a leader if its neighbors are only leaders and it coordinates with its neighbors to achieve a desired formation. A UAV is called a follower if it has at least one neighbor in the multi-UAV system and it coordinates with its neighbors to achieve the containment.

Assume that in multi-UAV system (2) there are M ($M < N$) followers with states $\phi_i(t)$ ($i = 1, 2, \dots, M$) and $N-M$ leaders with states $\phi_j(t)$ ($j = M+1, M+2, \dots, N$). Denote by $F = \{1, 2, \dots, M\}$ and $E = \{M+1, M+2, \dots, N\}$, the follower subscript set and leader subscript set, respectively. A time-varying formation for leaders is specified by a vector $h_E(t) = [h_{M+1}^T(t), h_{M+2}^T(t), \dots, h_N^T(t)]^T \in \mathbb{R}^{2(N-M)}$ with $h_i(t) = [h_{ix}(t), h_{iv}(t)]^T$ ($i \in E$) piecewise continuously differentiable.

Definition 2: Leaders in the multi-UAV system (2) are said to achieve time-varying formation $h_E(t)$ if there exists a vector-valued function $r(t) \in \mathbb{R}^2$, such that

$$\lim_{t \rightarrow \infty} (\phi_i(t) - h_i(t) - r(t)) = 0 \quad (i \in E) \quad (3)$$

where $r(t)$ is called a formation reference function.

In Definition 2, $h_i(t)$ ($i \in E$) is used to characterize the desired time-varying formation configuration and $r(t)$ is a representation of the macroscopic movement of the whole formation. To explain the roles of $h_i(t)$ and $r(t)$ more clearly, consider the following illustration example.

Illustration Example 1: Consider a multi-UAV system with four quadrotors in the XY plane. For simplicity, the states of the four quadrotors are required to achieve a time-invariant square formation with edge $\sqrt{2}l$. To specify the given formation shape, the vector $h_E = [h_1^T, h_2^T, h_3^T, h_4^T]^T$ can be chosen as $h_1 = [-l, 0]^T$, $h_2 = [0, l]^T$, $h_3 = [l, 0]^T$, and $h_4 = [0, -l]^T$. If Definition 2 is satisfied, it follows from (3) that $\lim_{t \rightarrow \infty} ((\phi_i(t) - \phi_j(t)) - (h_i - h_j)) = 0$ ($i, j = 1, 2, 3, 4$), which means that the two squares formed by h_i and $\phi_i(t)$ are congruent with each other. Therefore, the given square formation is achieved by the multi-UAV system. Fig. 1 depicts the geometric relationships of $\phi_i(t)$, h_i ($i = 1, 2, 3, 4$) and $r(t)$. From Fig. 1, one gets that the formation reference $r(t)$ can be used to describe the macroscopic movement of the whole formation, and h_i denotes the relative offset vector of $\phi_i(t)$ relative to $r(t)$.

Definition 3: Multi-UAV system (2) is said to achieve containment if for any $j \in F$, there exist nonnegative constants α_{jk} ($k \in E$) satisfying $\sum_{k=M+1}^N \alpha_{jk} = 1$, such that

$$\lim_{t \rightarrow \infty} \left(\phi_j(t) - \sum_{k=M+1}^N \alpha_{jk} \phi_k(t) \right) = 0. \quad (4)$$

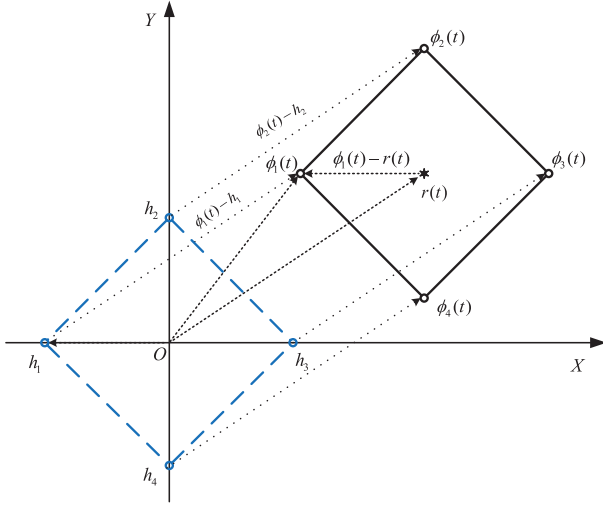


Fig. 1. Illustration example for square formation with four UAVs.

Definition 4: Multi-UAV system (2) is said to achieve formation-containment if there exist a vector-valued function $r(t) \in \mathbb{R}^2$ and nonnegative constants α_{jk} ($k \in E$) satisfying $\sum_{k=M+1}^N \alpha_{jk} = 1$, such that for any $i \in E$ and $j \in F$, (3) and (4) hold, respectively.

Remark 2: From Definitions 2–4, one sees that for multi-UAV system (2), if $M = 0$, formation-containment problems become the formation problems discussed in [18], [19], [22], [25], and [27]. If $h(t) \equiv 0$ and leaders have no neighbors (i.e., for all $i, j \in E$, $w_{ij} = 0$), the formation-containment problem becomes the containment problem discussed in [31]–[34]. Therefore, for multi-UAV system (2), formation problems and containment problems can be regarded as special cases of the formation-containment problems discussed in this paper.

Consider the following formation-containment protocols for multi-UAV system (2)

$$u_i(t) = K_1 \phi_i(t) + K_2 \sum_{j \in N_i} w_{ij} (\phi_i(t) - \phi_j(t)), \quad i \in F \quad (5)$$

$$u_i(t) = K_1 \phi_i(t) + K_3 \sum_{j \in N_i} w_{ij} ((\phi_i(t) - h_i(t)) - (\phi_j(t) - h_j(t))), \quad i \in E \quad (6)$$

where $K_l = [k_{l1}, k_{l2}]$ ($l = 1, 2, 3$) are constant gain matrices to be determined later.

Remark 3: In protocols (5) and (6), the gain matrix K_1 can be chosen to expand the feasible formation set or specify the motion modes of the formation reference. K_2 and K_3 can be used to drive the states of followers to converge to the convex hull formed by those of leaders and propel the states of leaders to achieve the desired time-varying formation, respectively. It should be pointed out that choosing $K_1 = 0$, protocols (5) and (6) become the ones using only relative information about neighbors.

In this paper, the following three problems for multi-UAV system (2) with protocols (6) and (7) are mainly studied: i) under what conditions formation-containment can be achieved; ii) how to design protocols (6) and (7) to

achieve formation-containment; and iii) how to demonstrate the theoretical results on practical quadrotor formation-containment platform.

III. FORMATION-CONTAINMENT ANALYSIS AND PROTOCOL DESIGN

In this section, first, sufficient conditions for multi-UAV system (2) under protocols (5) and (6) to achieve formation-containment are proposed. Then, an explicit expression to describe the relationship among the states of followers, the time-varying formation of the leaders, and the formation reference are derived. Finally, an approach to design the protocols (5) and (6) is presented.

Let G_E denote the interaction topology among the leaders. Under Definition 1, the Laplacian matrix corresponding to G has the following form:

$$L = \begin{bmatrix} L_1 & L_2 \\ 0 & L_3 \end{bmatrix}$$

where $L_1 \in \mathbb{R}^{M \times M}$, $L_2 \in \mathbb{R}^{M \times (N-M)}$, and $L_3 \in \mathbb{R}^{(N-M) \times (N-M)}$, which is the Laplacian matrix corresponding to G_E .

Assumption 1: The interaction topology among the leaders has a spanning tree.

Assumption 2: For each follower, there exists at least one leader that has a directed path to it.

Under Assumption 2, the following lemma can be obtained.

Lemma 2 [30]: If the directed interaction topology G satisfies Assumption 2, then all the eigenvalues of L_1 have positive real parts, each entry of $-L_1^{-1} L_2$ is nonnegative, and each row of $-L_1^{-1} L_2$ has a sum equal to one.

Let $\phi_F(t) = [\phi_1^T(t), \phi_2^T(t), \dots, \phi_M^T(t)]^T$, and $\phi_E(t) = [\phi_{M+1}^T(t), \phi_{M+2}^T(t), \dots, \phi_N^T(t)]^T$. Under protocols (5) and (6), multi-UAV system (2) can be written in a compact form as

$$\dot{\phi}_F(t) = (I_M \otimes (B_1 B_2^T + B_2 K_1) + L_1 \otimes B_2 K_2) \phi_F(t) + (L_2 \otimes B_2 K_2) \phi_E(t) \quad (7)$$

$$\dot{\phi}_E(t) = (I_{N-M} \otimes (B_1 B_2^T + B_2 K_1) + L_3 \otimes B_2 K_3) \phi_E(t) - (L_3 \otimes B_2 K_3) h_E(t). \quad (8)$$

Let λ_i ($i \in E$) be the eigenvalue of the matrix L_3 , where $\lambda_{M+1} = 0$ with the associated eigenvector $\bar{u}_{M+1} = \mathbf{1}$ and $0 < \text{Re}(\lambda_{M+2}) \leq \dots \leq \text{Re}(\lambda_N)$. There exists a nonsingular matrix $U_E = [\bar{u}_{M+1}, \bar{u}_{M+2}, \dots, \bar{u}_N]$ with $U_E^{-1} = [\tilde{u}_{M+1}^H, \tilde{u}_{M+2}^H, \dots, \tilde{u}_N^H]^H$, such that $U_E^{-1} L_3 U_E = J_E$, where J_E is the Jordan canonical form of L_3 . By Lemma 1, the structure of U_E and J_E has the form of $J_E = \text{diag}\{0, \bar{J}_E\}$, where $\bar{J}_E$ consists of Jordan blocks corresponding to λ_i ($i = M+2, M+3, \dots, N$). Let $\tilde{U}_E = [\tilde{u}_{M+2}^H, \tilde{u}_{M+3}^H, \dots, \tilde{u}_N^H]^H$. Define $U_F \in \mathbb{C}^{M \times M}$ as a nonsingular matrix, such that $U_F^{-1} L_1 U_F = \Lambda_F$, where Λ_F is an upper triangular matrix with diagonal entries λ_i ($i = 1, 2, \dots, M$) and $\text{Re}(\lambda_1) \leq \text{Re}(\lambda_2) \leq \dots \leq \text{Re}(\lambda_M)$.

Lemma 3 [45]: The system $\dot{X} = MX$, where M is a 2×2 complex matrix with characteristic polynomial $f(s) = s^2 + a_1 s + a_2$, is asymptotically stable if and only if $\text{Re}(a_1) > 0$ and $\text{Re}(a_1) \text{Re}(a_1 \bar{a}_2) - \text{Im}(a_2)^2 > 0$.

The following theorem gives sufficient conditions for multi-UAV system (2) under the proposed protocols to achieve formation-containment.

Theorem 1: Multi-UAV system (2) under protocols (5) and (6) achieves formation-containment if the following conditions hold simultaneously.

- (i) For any $i \in E$ and $j \in N_i$, the desired formation satisfies the following formation feasibility constraint:

$$\lim_{t \rightarrow \infty} ((B_1 B_2^T + B_2 K_1)(h_i(t) - h_j(t)) - (\dot{h}_i(t) - \dot{h}_j(t))) = 0. \quad (9)$$

- (ii) For any $i \in \{M+2, M+3, \dots, N\}$

$$\begin{cases} k_{12} + \operatorname{Re}(\lambda_i)k_{32} < 0 \\ (k_{12} + \operatorname{Re}(\lambda_i)k_{32})\Psi_i + \operatorname{Im}(\lambda_i)^2 k_{31}^2 < 0 \end{cases} \quad (10)$$

$$\text{where } \Psi_i = k_{12}k_{11} - \operatorname{Re}(\lambda_i)(k_{12}k_{31} + k_{11}k_{32}) + (\operatorname{Re}(\lambda_i)^2 + \operatorname{Im}(\lambda_i)^2)k_{31}k_{32}.$$

- (iii) For any $i \in F$

$$\begin{cases} k_{12} + \operatorname{Re}(\lambda_i)k_{22} < 0 \\ (k_{12} + \operatorname{Re}(\lambda_i)k_{22})\Psi_i + \operatorname{Im}(\lambda_i)^2 k_{21}^2 < 0 \end{cases} \quad (11)$$

$$\text{where } \Psi_i = k_{12}k_{11} - \operatorname{Re}(\lambda_i)(k_{12}k_{21} + k_{11}k_{22}) + (\operatorname{Re}(\lambda_i)^2 + \operatorname{Im}(\lambda_i)^2)k_{21}k_{22}.$$

Proof: Let $\bar{\phi}_i(t) = \phi_i(t) - h_i(t)$ ($i \in E$) and $\bar{\phi}_E(t) = [\bar{\phi}_{M+1}^T(t), \bar{\phi}_{M+2}^T(t), \dots, \bar{\phi}_N^T(t)]^T$. Then, the system (8) can be rewritten as

$$\begin{aligned} \dot{\bar{\phi}}_E(t) &= (I_{N-M} \otimes (B_1 B_2^T + B_2 K_1) + L_3 \otimes B_2 K_3) \bar{\phi}_E(t) \\ &\quad + (I_{N-M} \otimes (B_1 B_2^T + B_2 K_1)) h_E(t) \\ &\quad - (I_{N-M} \otimes I_2) \dot{h}_E(t). \end{aligned} \quad (12)$$

Let $\theta_E(t) = (\tilde{u}_{M+1} \otimes I_2) \bar{\phi}_E(t)$ and $\zeta_E(t) = (\tilde{U}_E \otimes I_2) \bar{\phi}_E(t)$, then the system (12) can be transformed into

$$\begin{aligned} \dot{\theta}_E(t) &= (B_1 B_2^T + B_2 K_1) \theta_E(t) \\ &\quad + (\tilde{u}_{M+1} \otimes (B_1 B_2^T + B_2 K_1)) h_E(t) \\ &\quad - (\tilde{u}_{M+1} \otimes I_2) \dot{h}_E(t) \end{aligned} \quad (13)$$

$$\begin{aligned} \dot{\zeta}_E(t) &= (I_{N-M-1} \otimes (B_1 B_2^T + B_2 K_1) + \bar{J}_E \otimes B_2 K_3) \zeta_E(t) \\ &\quad + (\tilde{U}_E \otimes (B_1 B_2^T + B_2 K_1)) h_E(t) - (\tilde{U}_E \otimes I_2) \dot{h}_E(t). \end{aligned} \quad (14)$$

If condition (i) holds, there is

$$\lim_{t \rightarrow \infty} ((L_3 \otimes (B_1 B_2^T + B_2 K_1)) h_E(t) - (L_3 \otimes I) \dot{h}_E(t)) = 0. \quad (15)$$

Substituting $L_3 = U_E J_E U_E^{-1}$ into (15) and then premultiplying the both sides of (15) by $U_E^{-1} \otimes I_2$ yield

$$\begin{aligned} \lim_{t \rightarrow \infty} ((\bar{J}_E \tilde{U}_E \otimes (B_1 B_2^T + B_2 K_1)) h_E(t) \\ - (\bar{J}_E \tilde{U}_E \otimes I_2) \dot{h}_E(t)) = 0. \end{aligned} \quad (16)$$

Because G_E has a spanning tree, by Lemma 1 and the structure of J_E , $\bar{J}_E$ is nonsingular. Premultiplying the both sides of (16) by $\bar{J}_E^{-1} \otimes I_2$, one has

$$\lim_{t \rightarrow \infty} ((\tilde{U}_E \otimes (B_1 B_2^T + B_2 K_1)) h_E(t) - (\tilde{U}_E \otimes I_2) \dot{h}_E(t)) = 0. \quad (17)$$

Consider the following $N - M - 1$ subsystems

$$\dot{\zeta}_i(t) = (B_1 B_2^T + B_2 K_1 + \lambda_i B_2 K_3) \zeta_i(t) \quad (18)$$

where $i = M+2, M+3, \dots, N$. The characteristic polynomial of subsystems (18) are $f_i(s) = s^2 - (k_{12} + \lambda_i k_{32})s - (k_{11} + \lambda_i k_{31})$. If condition (ii) holds, by Lemma 3, it can be obtained that the $N - M - 1$ subsystems described by (18) are asymptotically stable. Based on (18) and the structure of $\bar{J}_E$, it holds that the system described by

$$\dot{\zeta}_E(t) = (I_{N-M-1} \otimes (B_1 B_2^T + B_2 K_1) + \bar{J}_E \otimes B_2 K_3) \zeta_E(t) \quad (19)$$

is asymptotically stable. From (14), (17), and (19), one gets that

$$\lim_{t \rightarrow \infty} \zeta_E(t) = 0. \quad (20)$$

Define

$$\bar{\phi}_{EC}(t) = \mathbf{1} \otimes \theta_E(t) \quad (21)$$

$$\bar{\phi}_{E\bar{C}}(t) = \bar{\phi}_E(t) - \bar{\phi}_{EC}(t). \quad (22)$$

Let $e_1 \in \mathbb{R}^{N-M}$ be a vector with 1 as its first component and 0 elsewhere. Because $[\theta_E^H(t), 0]^H = e_1 \otimes \theta_E(t)$ and $U_E e_1 = \mathbf{1}$, one has

$$\bar{\phi}_{EC}(t) = (U_E \otimes I) [\theta_E^H(t), 0]^H. \quad (23)$$

Note that $\bar{\phi}_E(t) = (U_E \otimes I) [\theta_E^H(t), \zeta_E^H(t)]^H$. From (21) and (22), it holds that

$$\bar{\phi}_{E\bar{C}}(t) = (U_E \otimes I) [0, \zeta_E^H(t)]^H. \quad (24)$$

Since $U_E \otimes I$ is nonsingular, from (23) and (24), one gets that $\bar{\phi}_{EC}(t)$ and $\bar{\phi}_{E\bar{C}}(t)$ are linearly independent. Therefore, it follows from (20), (22), and (24) that:

$$\lim_{t \rightarrow \infty} (\bar{\phi}_E(t) - \mathbf{1} \otimes \theta_E(t)) = 0 \quad (25)$$

that is

$$\lim_{t \rightarrow \infty} (\phi_i(t) - h_i(t) - \theta_E(t)) = 0 \quad (i \in E). \quad (26)$$

From (26), one gets that multi-UAV system (8) achieves the predefined time-varying formation specified by $h_E(t)$.

Let

$$\varphi_i(t) = \sum_{j \in N_i} w_{ij} (\phi_i(t) - \phi_j(t)) \quad (i \in F)$$

and $\varphi_F(t) = [\varphi_1^T(t), \varphi_2^T(t), \dots, \varphi_M^T(t)]^T$, then one gets

$$\varphi_F(t) = (L_1 \otimes I_2) \phi_F(t) + (L_2 \otimes I_2) \phi_E(t). \quad (27)$$

From (7), (8), and (27), it can be obtained that

$$\begin{aligned} \dot{\phi}_F(t) &= (I_M \otimes (B_1 B_2^T + B_2 K_1) + (L_1 \otimes B_2 K_2)) \varphi_F(t) \\ &\quad + ((L_2 L_3) \otimes (B_2 K_3)) (\phi_E(t) - h_E(t)). \end{aligned} \quad (28)$$

Let $\bar{\varphi}_F(t) = (U_F^{-1} \otimes I) \varphi_F(t) = [\bar{\varphi}_1^H(t), \bar{\varphi}_2^H(t), \dots, \bar{\varphi}_M^H(t)]^H$. Then, system (28) can be transformed into

$$\begin{aligned} \dot{\bar{\varphi}}_F(t) &= (I_M \otimes (B_1 B_2^T + B_2 K_1) + (\Lambda_F \otimes B_2 K_2)) \bar{\varphi}_F(t) \\ &\quad + ((U_F^{-1} L_2 L_3) \otimes (B_2 K_3)) (\phi_E(t) - h_E(t)). \end{aligned} \quad (29)$$

When the leaders achieve time-varying formation $h_E(t)$, there is

$$\lim_{t \rightarrow \infty} (L_3 \otimes (B_2 K_3))(\phi_E(t) - h_E(t)) = \lim_{t \rightarrow \infty} (L_3 \otimes (B_2 K_3))(\mathbf{1} \otimes r(t)). \quad (30)$$

Since $L_3 \mathbf{1} = 0$, one has

$$\lim_{t \rightarrow \infty} ((U_F^{-1} L_2 L_3) \otimes (B_2 K_3))(\phi_E(t) - h_E(t)) = 0. \quad (31)$$

With a similar analysis as for system (19), it can be verified that if condition (iii) holds, then $I_M \otimes (B_1 B_2^T + B_2 K_1) + (\Lambda_F \otimes B_2 K_2)$ is Hurwitz. Therefore, $\lim_{t \rightarrow \infty} \bar{\phi}_F(t) = 0$. Since U_F is nonsingular, one has

$$\lim_{t \rightarrow \infty} \phi_F(t) = 0. \quad (32)$$

It follows from (27) and (32) that

$$\lim_{t \rightarrow \infty} (\phi_F(t) - (-L_1^{-1} L_2 \otimes I_2) \phi_E(t)) = 0. \quad (33)$$

From Lemma 2, (26), and (33), one gets that multi-UAV system (2) under protocols (5) and (6) achieves formation-containment. The proof of Theorem 1 is completed. ■

Remark 4: In the theoretical analysis, for simplicity of description, the dimension n is assumed to be 1. It should be pointed out that all the results in Theorem 1 can be extended to the higher dimensional space directly by using the Kronecker product. Theorem 1 presents sufficient conditions for multi-UAV system (2) with directed topologies under protocols (5) and (6) to achieve formation-containment. Note that the dynamics of each agent in [37]–[40] is first order or the topology is undirected. From (9), one gets that not all the formation can be achieved by multi-UAV system (2), and the desired formation should satisfy the formation feasibility constraint (9) determined by the dynamics of each UAV. Moreover, the application of K_1 can expand the feasible formation set.

Remark 5: It should be pointed out that the formation reference function $r(t)$ is a representation of the macroscopic movement of the whole formation-containment. From (3) and (26), one gets that $\lim_{t \rightarrow \infty} (\theta_E(t) - r(t)) = 0$, which means that the formation reference is determined by $\theta_E(t)$. By solving the differential equation (13), an explicit expression of the formation reference $r(t)$ can be obtained. Moreover, from (13), one gets that the motion modes of the formation reference can be specified by assigning the eigenvalues of $B_1 B_2^T + B_2 K_1$ at the desired locations in the complex plane using K_1 .

It is proven that the states of followers can converge to the convex hull formed by those of leaders in Theorem 1. Furthermore, the following theorem shows the explicit relationship among the states of followers, the time-varying formation of the leaders and the formation reference.

Theorem 2: If conditions (i)–(iii) in Theorem 1 are satisfied, multi-UAV system (2) under protocols (5) and (6) achieves formation-containment, and the states of followers satisfy

$$\lim_{t \rightarrow \infty} \left(\phi_i(t) - \sum_{j=M+1}^N l_{ij} h_j(t) - \theta_E(t) \right) = 0 \quad (34)$$

where $i \in F$ and l_{ij} is the entries of $-L_1^{-1} L_2$.

Proof: If conditions (i), (ii) and (iii) in Theorem 1 are satisfied, it holds that multi-UAV system (2) under protocols (5) and (6) achieves formation-containment. From the proof of Theorem 1, one gets that (25) and (33) hold. From (25) and (33), one has

$$\lim_{t \rightarrow \infty} (\phi_F(t) - (-L_1^{-1} L_2 \otimes I_2)(h_E(t) + \mathbf{1} \otimes \theta_E(t))) = 0. \quad (35)$$

It follows from Lemma 2 that

$$-L_1^{-1} L_2 \mathbf{1} = \mathbf{1}. \quad (36)$$

From (35) and (36), it can be obtained that

$$\lim_{t \rightarrow \infty} (\phi_F(t) - (-L_1^{-1} L_2 \otimes I_2) h_E(t) - \mathbf{1} \otimes \theta_E(t)) = 0 \quad (37)$$

which means that (34) holds. This completes the proof. ■

Remark 6: Theorem 2 shows that the states of followers are jointly determined by the topology, the time-varying formation $h_E(t)$, and the formation reference. From (34), one gets that the states of followers keep a time-varying formation specified by the convex combination of the formation $h_E(t)$ for the leaders with respect to $r(t)$. Therefore, the states of followers not only converge to the convex hull formed by those of leaders but also keep certain formation specified by the convex combination of the formation for the leaders.

From Theorems 1 and 2, one gets that the state $\theta_E(t)$ of subsystem (13) determines the formation reference function $r(t)$ that represents the macroscopic movement of the whole formation-containment. Gain matrix K_1 can be chosen to satisfy the formation feasibility constraint (9) or specify the motion modes of the formation reference $r(t)$ by assigning the eigenvalues of $B_1 B_2^T + B_2 K_1$ at the desired locations in the complex plane. After K_1 is chosen, the following theorem provides an approach to determine the gain matrices K_2 and K_3 .

Theorem 3: If condition (i) in Theorem 1 holds, then multi-UAV system (2) achieves formation-containment by protocols (5) and (6) with $K_2 = -\alpha[\text{Re}(\lambda_1)]^{-1} R^{-1} B_2^T P$ and $K_3 = -\beta[\text{Re}(\lambda_{M+2})]^{-1} R^{-1} B_2^T P$, where $\alpha > 0.5$, $\beta > 0.5$ are given constants, and P is the positive definite solution to the algebraic Riccati equation

$$P(B_1 B_2^T + B_2 K_1) + (B_1 B_2^T + B_2 K_1)^T P - P B_2 R^{-1} B_2^T P + I = 0 \quad (38)$$

with $R^T = R > 0$.

Proof: Since $(B_1 B_2^T, B_2)$ is stabilizable and $(I, B_1 B_2^T)$ is observable, for any given $R^T = R > 0$, algebraic Riccati equation (38) has a unique solution $P^T = P > 0$. First, consider the stabilities of the $N - M - 1$ subsystems described by (18). Construct the following Lyapunov function candidates:

$$V_i(t) = \bar{\zeta}_i^H(t) P \bar{\zeta}_i(t) \quad (i = M + 2, M + 3, \dots, N). \quad (39)$$

Taking the derivative of $V_i(t)$ along the trajectories of subsystems (18) gives

$$\begin{aligned} \dot{V}_i(t) &= \bar{\zeta}_i^H(t) \left((B_1 B_2^T + B_2 K_1)^T P + P (B_1 B_2^T + B_2 K_1) \right) \bar{\zeta}_i(t) \\ &\quad + \bar{\zeta}_i^H(t) \left(\lambda_i^H (B_2 K_3)^T P + \lambda_i P B_2 K_3 \right) \bar{\zeta}_i(t). \end{aligned} \quad (40)$$

Substituting $K_3 = -\beta [\text{Re}(\lambda_{M+2})]^{-1} R^{-1} B_2^T P$ and

$$\begin{aligned} P(B_1 B_2^T + B_2 K_1) + (B_1 B_2^T + B_2 K_1)^T P \\ = P B_2 R^{-1} B_2^T P - I \end{aligned}$$

into (40) one has

$$\begin{aligned} \dot{V}_i(t) &= -\bar{\zeta}_i^H(t) \bar{\zeta}_i(t) + (1 - 2\beta [\text{Re}(\lambda_{M+2})]^{-1} \text{Re}(\lambda_i)) \\ &\quad \times \bar{\zeta}_i^H(t) (P B_2 R^{-1} B_2^T P) \bar{\zeta}_i(t). \end{aligned} \quad (41)$$

Since $1 - 2\beta [\text{Re}(\lambda_{M+2})]^{-1} \text{Re}(\lambda_i) < 0$, it follows from (41) that $\lim_{t \rightarrow \infty} \bar{\zeta}_i(t) = 0$, which means that the $N - M - 1$ subsystems described by (18) are asymptotically stable. Since condition (i) is satisfied, one gets $\lim_{t \rightarrow \infty} \zeta_E(t) = 0$. With a similar analysis as for subsystems (18), it can be verified that $K_2 = -\alpha [\text{Re}(\lambda_1)]^{-1} R^{-1} B_2^T P$ can guarantee that $B_1 B_2^T + B_2 K_1 + \lambda_i B_2 K_2$ ($i \in F$) is Hurwitz, which means that $I_M \otimes (B_1 B_2^T + B_2 K_1) + (\Lambda_F \otimes B_2 K_2)$ is Hurwitz. From the proof of Theorem 1, one can conclude that multi-UAV system (2) achieves formation-containment by protocols (5) and (6). This completes the proof. ■

Remark 7: From (38), one sees that to determine the gain matrices K_2 and K_3 , only an algebraic Riccati equation needs to be solved. Moreover, the existence of K_2 and K_3 can be guaranteed. Therefore, the approach to design the formation-containment protocols (5) and (6) is quite practical.

IV. SIMULATION AND EXPERIMENTAL RESULTS

In this section, first, a quadrotor formation-containment platform with five quadrotors is introduced. Then both a numerical simulation and a practical experiment are carried out on the quadrotor formation-containment platform with five quadrotors in Example 1. Moreover, to demonstrate the scalability of the obtained results, a simulation result with six leaders and eight followers in the 3-D space is given as Example 2. Due to the quantity limitation of the quadrotor UAVs in our laboratory, the experimental results are not presented in Example 2.

The quadrotor formation-containment platform consists of five quadrotors equipped with onboard flight control systems (FCSs) and a ground control station (GCS), as shown in Fig. 2. The hardware of the quadrotor is developed based on the frame of the commercial X650V from Xaircraft [46] with a tip-to-tip wingspan of 65 cm, a battery life of 12 min, and a maximum takeoff weight of 1800 g. The FCS is developed based on a TMS320F28335 DSP with a clock frequency of 135 MHz. The attitude and acceleration of the quadrotor are estimated by three one-axis gyroscopes, a three-axis accelerometer, and a three-axis magnetometer. A GPS module with an accuracy of 1.2-m circular error probable is employed to obtain the position and velocity of the quadrotor with an output rate of 10 Hz. The height of the quadrotor



Fig. 2. Quadrotor formation-containment platform.

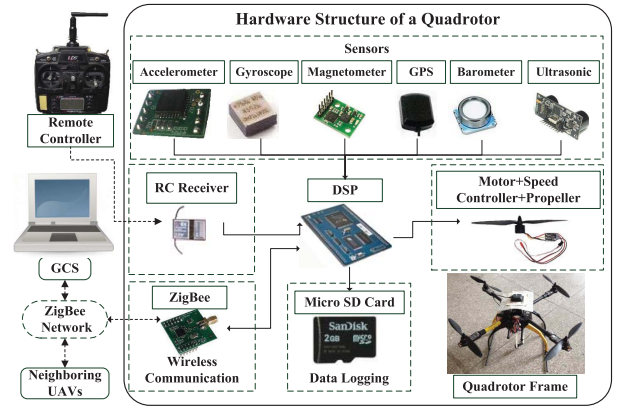


Fig. 3. Hardware structure of the quadrotor formation-containment platform.

is measured by an ultrasonic range finder for the near the ground case or a barometer for the far away from the ground case. A 2-GB micro SD card is mounted onboard for data logging. Zigbee modules are employed in each quadrotor and the GCS for data transmission. An RC receiver is kept in each quadrotor to deal with the emergency situation. The GCS is developed based on Labview 8.2.1 in the Windows XP environment by which the flight parameters of all quadrotors can be monitored and control commands can be sent to quadrotors. Fig. 3 describes the hardware structure of the quadrotor formation-containment platform.

Example 1 (Simulation and Experiment with Five Quadrotors): The formation-containment control is implemented in the horizontal plane ($n = 2$), i.e., movements of followers and leaders along the X - and Y -axes are controlled by the formation-containment protocols (5) and (6), respectively, with a rate of 5 Hz. The height of each quadrotor is specified to be constant. The pitch, roll, and yaw angles of each quadrotor are controlled by three decoupled proportion differentiation controllers shown in [46] with a rate of 500 Hz, respectively, as the inner loop. Using the Kronecker product, the dynamics of the quadrotor swarm system in the two-dimensional space can be described by (2) with $\phi_i(t) = [x_{iX}(t), v_{iX}(t), x_{iY}(t), v_{iY}(t)]^T$, $u_i(t) = [u_{iX}(t), u_{iY}(t)]^T$ ($i = 1, 2, \dots, N$), $B_1 = I_2 \otimes [1, 0]^T$, $B_2 = I_2 \otimes [0, 1]^T$, and $h_i(t) = [h_{iX}(t), h_{iY}(t)]^T$ ($i \in E$).

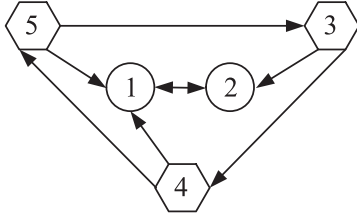
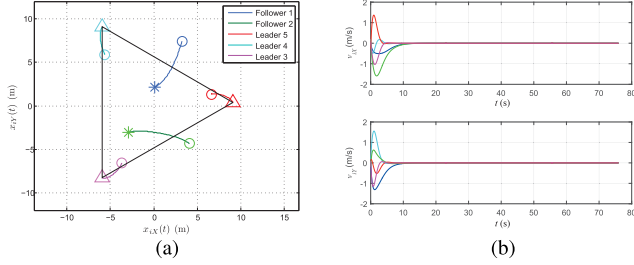
Fig. 4. Directed interaction topology G_1 .

Fig. 5. Position and velocity trajectories for five quadrotors in simulation. (a) Positions. (b) Velocities.

Quadrotors 1 and 2 are assigned as followers, while quadrotors 3–5 are specified to be leaders, i.e., $M = 2$ and $N = 5$. The directed interaction topology of the quadrotor swarm system is shown in Fig. 4. For simplicity, the interaction topology is 0–1 weighted.

The three leaders are required to achieve a formation specified by

$$h_3(t) = \begin{bmatrix} 10 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad h_4(t) = \begin{bmatrix} 10 \cos(2\pi/3) \\ 0 \\ 10 \sin(2\pi/3) \\ 0 \end{bmatrix}$$

$$h_5(t) = \begin{bmatrix} 10 \cos(4\pi/3) \\ 0 \\ 10 \sin(4\pi/3) \\ 0 \end{bmatrix}.$$

If $h_i(t)$ ($i = 3, 4, 5$) is achieved, then the positions of three leaders keep a regular triangle with edge $10\sqrt{3}$ m while the velocities converge to a common value. Moreover, the positions of the two followers are required to converge to the triangle formation formed by those of the leaders, and the velocities of the followers reach an agreement with those of the leaders.

Due to the limitation of flight space and the requirement of performing the experiment within a visual range, the motion modes of the formation reference $r(t)$ are designed to be stable by choosing $K_1 = I_2 \otimes [0, -0.6]$ to specify the eigenvalues of $B_1 B_2^T + B_2 K_1$ at 0, 0, -0.6 , and -0.6 . In this configuration, when the desired formation-containment is achieved, the five quadrotors will be stationary. Let $K_2 = I_2 \otimes [-0.5, -0.8]$ and $K_3 = I_2 \otimes [-0.8, -0.8]$. It can be verified that the conditions in Theorem 1 are satisfied. Choose the initial states for all the quadrotors as $\phi_1(0) = [3.23, -0.04, 7.38, -0.01]^T$, $\phi_2(0) = [4.12, -0.01, -4.34, -0.19]^T$, $\phi_3(0) = [6.62, -0.26, 1.32, 0.13]^T$, $\phi_4(0) = [-5.66, 0.20, 5.86, 0.14]^T$, and

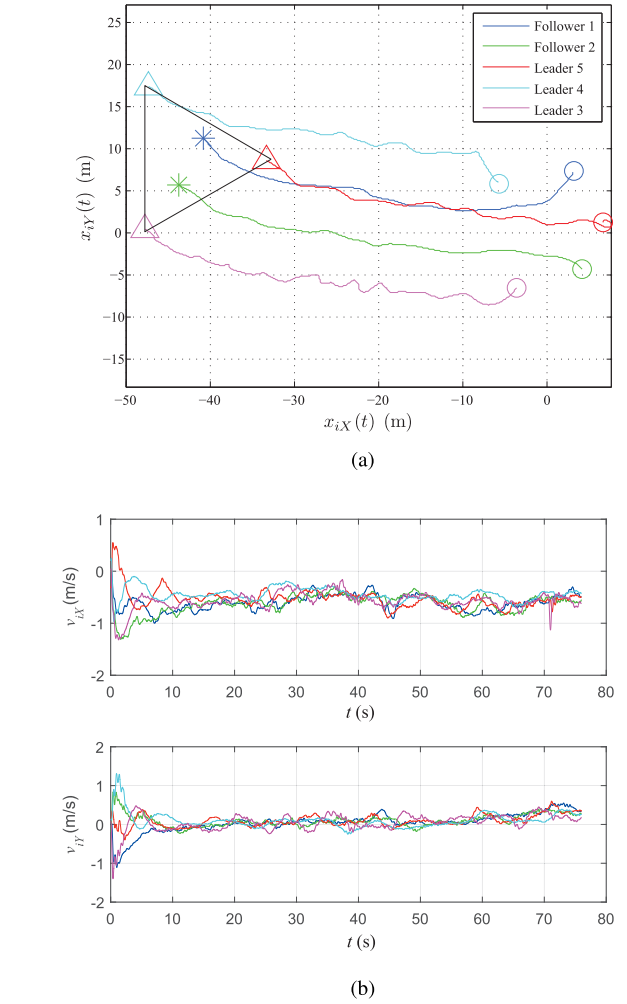


Fig. 6. Position and velocity trajectories for five quadrotors in experiment. (a) Positions. (b) Velocities.



Fig. 7. Formation-containment image for five quadrotors in experiment.

$\phi_5(0) = [-3.66, 0.04, -6.57, 0.08]^T$. Figs. 5 and 6 show the position and velocity trajectories of the five quadrotors in the simulation and experiment within 75 s, where the initial states of the quadrotors are marked by “o” and the final states of leaders and followers are denoted by “Δ” and “*,” respectively. Moreover, the convex hull formed by leaders is marked by solid lines. Fig. 7 shows an image of the formation-containment. The control inputs of the five quadrotors are shown in Fig. 8. Let the formation error $e_E(t) = \phi_E(t) - h_E(t) - \mathbf{1} \otimes \theta_E(t)$ and the containment

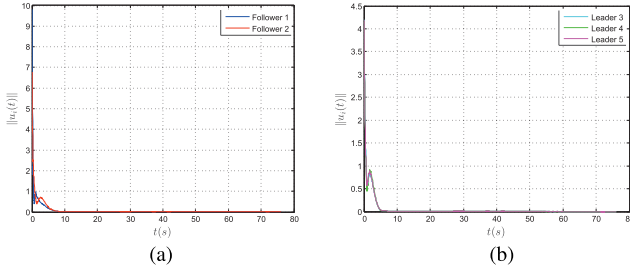


Fig. 8. Control inputs of the five quadrotors. (a) Followers. (b) Leaders.

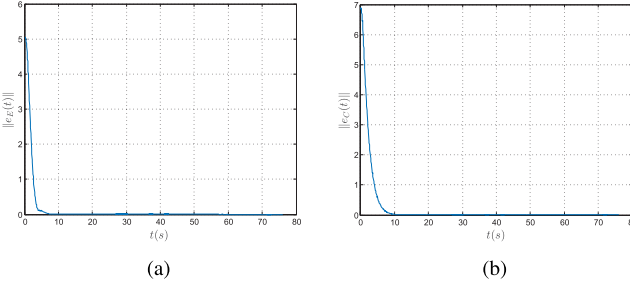


Fig. 9. Curves of the formation error for the leader quadrotors and the containment error for the follower quadrotors in simulation. (a) Formation error. (b) Containment error.

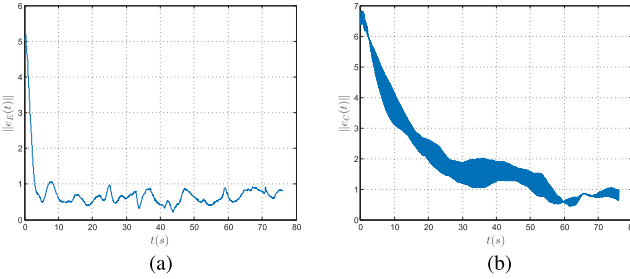
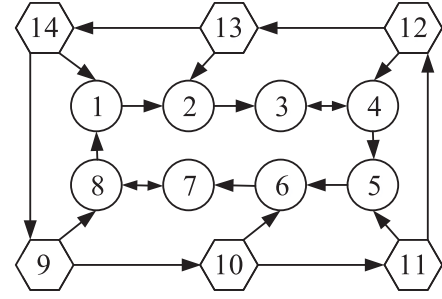


Fig. 10. Curves of the formation error for the leader quadrotors and the containment error for the follower quadrotors in experiment. (a) Formation error. (b) Containment error.

error $e_c(t) = \phi_F(t) - (-L_1^{-1}L_2 \otimes I_6)\phi_E(t)$. The curves of the formation error $e_E(t)$ for the leader quadrotors and the containment error $e_C(t)$ for the follower quadrotors in simulation and experiment are depicted in Figs. 9 and 10. From Figs. 5(a) and 6(a), one sees that the positions of leaders keep the desired regular triangle formation and the positions of followers stay in the convex hull formed by those of the leaders in both the simulation and the experiment. Figs. 5(b) and 6(b) show that all the velocities of the quadrotors reach an agreement. Therefore, the predefined formation-containment for the five quadrotors are achieved in both the simulation and the experiment. It should be pointed out that due to the existence of wind in the experiment, there is a drift in the position of the five quadrotors. However, the formation-containment is still realized under the disturbance of the wind. The video of the experiment can be found at http://v.youku.com/v_show/id_XNzA5ODc3NjA4.html. or <https://www.youtube.com/watch?v=-amro6wFOxY>.

Example 2 (Simulation with 14 Quadrotors): Assume that there are eight followers and six leaders in the

Fig. 11. Directed interaction topology G_2 .

quadrotor swarm system with directed interaction topology G_2 , as shown in Fig. 11. These agents move in the 3-D space (i.e., the XYZ space). Using the Kronecker product, the dynamics of the quadrotor swarm system in the 3-D space can be described by (2) with $\phi_i(t) = [x_{iX}(t), v_{iX}(t), x_{iY}(t), v_{iY}(t), x_{iZ}(t), v_{iZ}(t)]^T$, $u_i(t) = [u_{iX}(t), u_{iY}(t), u_{iZ}(t)]^T$ ($i = 1, 2, \dots, N$), $B_1 = I_3 \otimes [1, 0]^T$, $B_2 = I_3 \otimes [0, 1]^T$ and $h_i(t) = [h_{iX}(t), h_{iY}(t), h_{iX}(t), h_{iY}(t), h_{iX}(t), h_{iY}(t)]^T$ ($i \in E$). The leaders are required to achieve the following time-varying formation:

$$h_i(t) = \begin{bmatrix} 50 \cos(0.5t + (i-9)\pi/3) \\ -25 \sin(0.5t + (i-9)\pi/3) \\ 25\sqrt{2} \sin(0.5t + (i-9)\pi/3) \\ 12.5\sqrt{2} \cos(0.5t + (i-9)\pi/3) \\ -25\sqrt{2} \sin(0.5t + (i-9)\pi/3) \\ -12.5\sqrt{2} \cos(0.5t + (i-9)\pi/3) \end{bmatrix} \quad (i = 9, 10, \dots, 14).$$

From $h_E(t)$, one gets that if the predefined formation is realized, both the positions and velocities of the six leaders will keep a regular hexagon formation while keep rotating around the formation reference with a velocity of 0.5 rad/s. The motion modes of the formation reference can be designed to be oscillating by choosing $K_1 = I_3 \otimes [-0.25, 0]$ to specify the eigenvalues of $B_1 B_2^T + B_2 K_1$ at $0.5j, 0.5j, 0.5j, -0.5j, -0.5j$, and $-0.5j$ ($j^2 = -1$). In this case, the whole formation-containment will move periodically. It can be verified that condition (i) in Theorem 1 is satisfied. Choose $\alpha = 0.6$ and $\beta = 0.55$. Using the approach in Theorem 3, K_2 and K_3 can be obtained as $K_2 = I_3 \otimes [-0.6435, -1.3191]$ and $K_3 = I_3 \otimes [-0.8589, -1.7605]$. For simplicity, the initial position components and velocity components of each UAV are given by $30(\Theta - 0.5)$, where Θ is a pseudorandom value distributed in $(0, 1)$.

Fig. 12 shows the numerical simulation results of the 14 quadrotors within $t = 80$ s, where the initial states are marked by “o,” the final states of followers are represented by “*,” the final states of the leaders are denoted by “•,” “△,” “◇,” “□,” “×,” and hexagram, respectively, and the final states of the formation reference are marked by the pentagram. Fig. 13 shows the snapshots of position and velocity trajectories at $t = 70$ s. Fig. 14 gives the curves of the formation error $e_E(t)$ and the containment

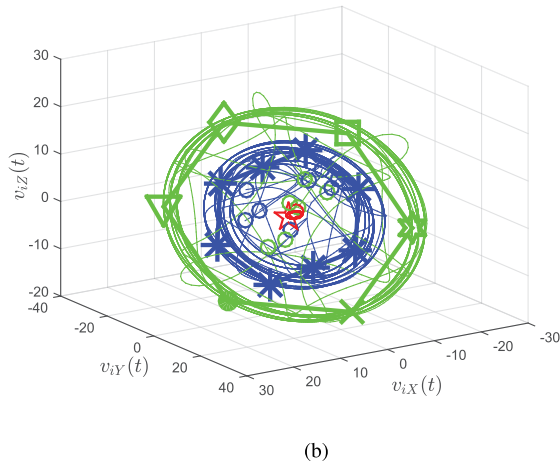
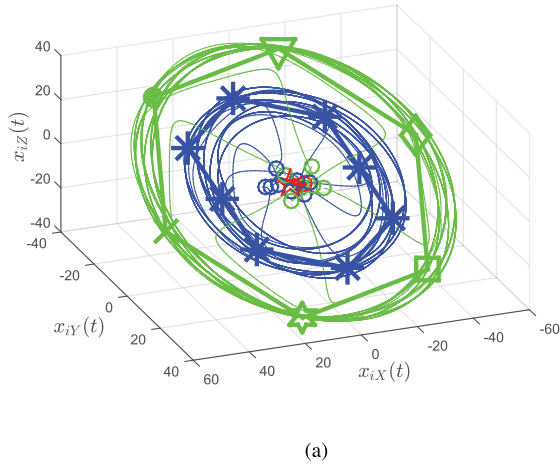


Fig. 12. Position and velocity trajectories for 14 quadrotors within $t = 80$ s. (a) Positions. (b) Velocities.

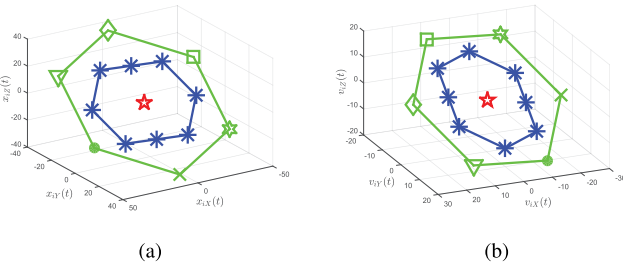


Fig. 13. Snapshots of position and velocity trajectories at $t = 70$ s. (a) Positions. (b) Velocities.

error $e_C(t)$. From Figs. 12–14, the following phenomena can be found: 1) both the position and velocity components of leaders form the regular hexagon formation; 2) the regular hexagon formation keeps rotating around the oscillating formation reference; and 3) the states of followers not only converge to the convex hull spanned by those of the leaders but also form hexagon formation. Therefore, the desired formation-containment is achieved. A video for the simulation with 14 quadrotors can be found at http://v.youku.com/v_show/id_XMzA2NDZOTA4OA==.html or <https://youtu.be/nV6zAwHS2wM>.

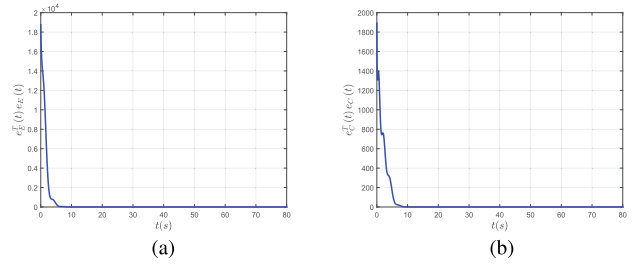


Fig. 14. Curves of the formation error for the leaders and the containment error for the followers. (a) Formation error. (b) Containment error.

V. CONCLUSION

Formation-containment control problems for multi-UAV systems with directed topologies were investigated. Sufficient conditions for multi-UAV systems to achieve formation-containment were presented. It was proven that the states of followers not only converge to the convex hull formed by those of leaders but also keep certain formation specified by the convex combination of the formation for the leaders. Approaches to determine the gain matrices of the formation-containment protocol were proposed by solving an algebraic Riccati equation. A formation-containment platform with five quadrotor UAVs was introduced, and both the simulation and experimental results were presented to demonstrate the effectiveness of the obtained results. An interesting topic for future research is to deal with the finite-time formation-containment control problems for multi-UAV systems.

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